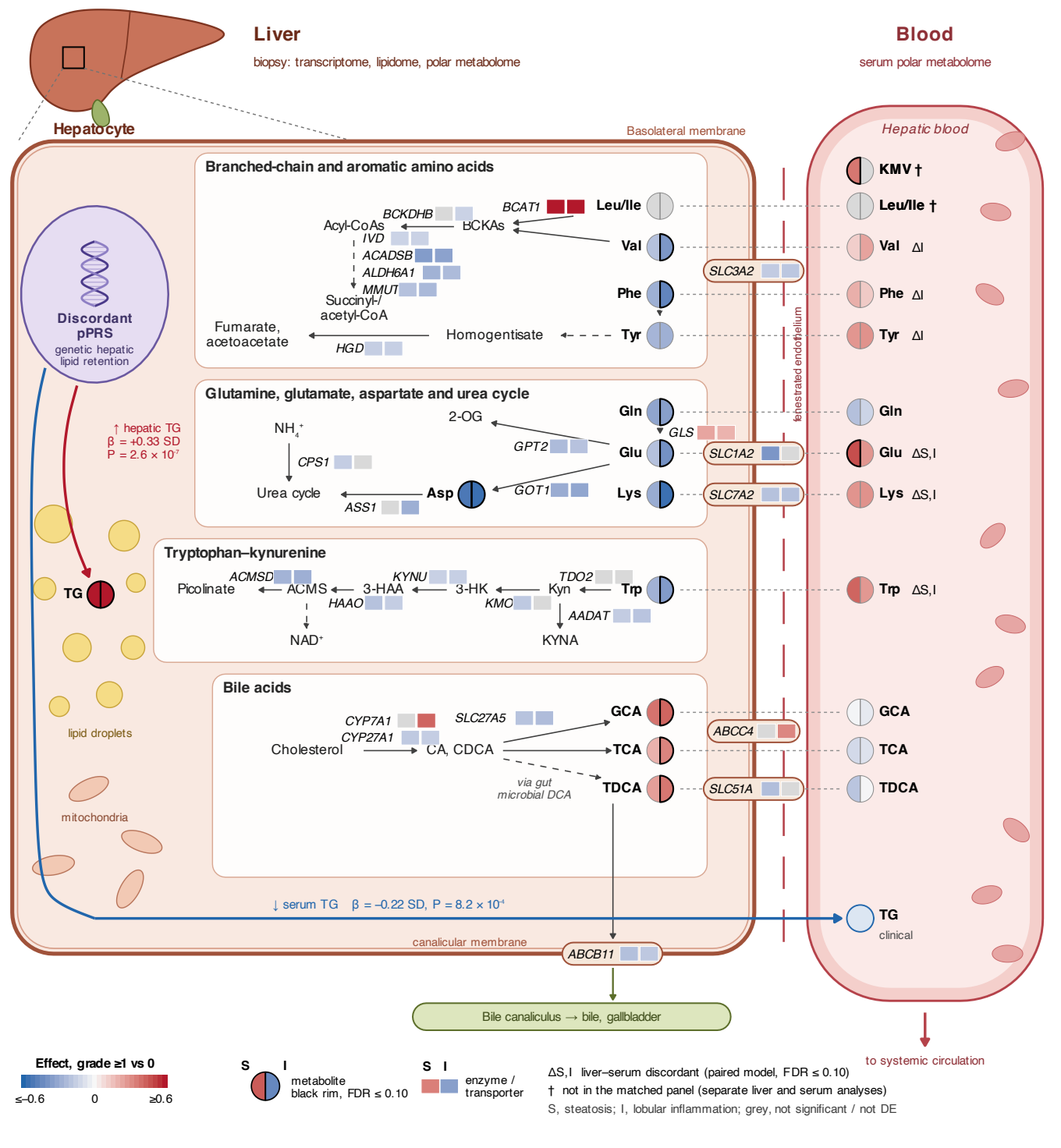

Liver
biopsy: transcriptome, lipidome, polar metabolome
Blood
serum polar metabolome
Hepatocyte
Basolateral membrane
Hepatic blood
Branched-chain and aromatic amino acids
BCKDHB
BCAT1
Leu/Ile
Acyl-CoAs
BCKAs
IVD
Val
ACADSB
ALDH6A1
MMUT
Phe
Succinyl-/ acetyl-CoA
Fumarate, acetoacetate
Homogentisate
HGD
Tyr
SLC3A2
KMV †
Leu/Ile †
Val ΔI
Phe ΔI
Tyr ΔI
Discordant pPRS
genetic hepatic lipid retention
↑ hepatic TG
β = +0.33 SD
P = 2.6 × 10⁻⁷
Glutamine, glutamate, aspartate and urea cycle
2-OG
Gln
GLS
NH4+
GPT2
Glu
CPS1
SLC1A2
Urea cycle
Asp
GOT1
Lys
SLC7A2
fenestrated endothelium
Gln
Glu ΔS,I
Lys ΔS,I
Tryptophan–kynurenine
ACMSD
KYNU
TDO2
Picolinate
ACMS
3-HAA
3-HK
Kyn
Trp
HAAO
KMO
AADAT
NAD+
KYNA
Trp ΔS,I
TG
lipid droplets
mitochondria
Bile acids
CYP7A1
SLC27A5
GCA
CYP27A1
Cholesterol
CA, CDCA
TCA
via gut microbial DCA
TDCA
ABCC4
SLC51A
GCA
TCA
TDCA
↓ serum TG β = –0.22 SD, P = 8.2 × 10⁻⁴
TG
clinical
canalicular membrane
ABCB11
Bile canaliculus → bile, gallbladder
to systemic circulation
Effect, grade ≥1 vs 0
≤–0.6
0
≥0.6
S I
metabolite
black rim, FDR ≤ 0.10
S I
enzyme / transporter
ΔS,I liver–serum discordant (paired model, FDR ≤ 0.10)
† not in the matched panel (separate liver and serum analyses)
S, steatosis; I, lobular inflammation; grey, not significant / not DE